

Energy: PV, Storage, Generator

Carpathian OER Commons — Utilities Module

Michel Garand | 2026-10-03 | Version v0.1 | CC BY-SA 4.0

Maturity: L1 — designed and costed, untested. Every cost line is the working paper's estimate for autumn 2026, labelled as such in Section 6; the loads are the paper's, recalculated at capacity, until the appliances are chosen. Nothing has been built. Read every figure below at this level and no higher.

Governed by Section 2 of the compendium. No child appears in this module in any form. The site is not named; site plans are schematic; photographs show structures and systems only. Design loads are site capacity, used for sizing only.

What keeps people safe is described in standing format. The shelter, the safety link, the battery, the generator, and the fuel store appear here as what each must do and how it is sized — never where it stands, what fuel is held, or when anything is run. The shelter's own supply is given as a principle and a method, without its figures.

Converted from the energy working paper, EN v0.1, Sections 1 to 4, 6, 7, and 9. Its Section 5 and 7.6 become the micro-hydro module; its Section 8 becomes the wood heat and hot water module. Five things change from the paper, decided on 2026-10-03: every load is recalculated at 34 people, not 25, and the loads the published modules add are carried; the December sun is taken from the building-climate standard, about half the paper's; the battery is four modules, not three, so that four days of the essential circuit still hold; the generator stands outdoors and its fuel apart, as the fire rules have held since 2024; and the second circuit is the general circuit.

Permits and local rules. This module is a guide, used in different regions and countries under different laws and guidelines. Depending on where it is built, at what scale, and for whom, permits, registrations, or approvals may be required — for building, water, sanitation, fire, waste, food, or the use of forest and land. Section 10 records what was found for Ukraine, dated, and unchecked until signed; a reader elsewhere checks their own rules with the competent authorities before building, and nothing here replaces that.

1. Purpose

Electricity for light, cold, communications, water treatment, and the safety of the site, through the winter and through the outages the country lives with, whether or not a grid connection exists. Panels, a battery, and a generator held in reserve, in standard parts that local trades sell, fit, and repair.

Five rules. Two circuits: an essential circuit — light, communications, the safety link, medical cold, the shelter, water treatment, the toilets' fans — sized to run for days without sun or fuel, and a general circuit that may go dark without harm. Storage before generation: four days of the essential circuit in the battery are worth more than more panels. Generation ranked by season: the sun carries the site from March to October; in winter the general circuit leans on the generator, or on a stream if one is ever developed. No

heat from electricity: heat and hot water come from wood and the sun, in the wood heat module. Nothing critical waits on fuel: the essential circuit runs on the battery and the sun alone; the generator only shortens the winter deficit of the general circuit.

Connections. Depends on: the appliance list, once the kitchen, medical room, and workshop are equipped; the Phase 1 walk, with the December horizon, the roof or ground for the array, the battery room, and the poorer ground; whether a grid connection exists; the electrician who will certify the installation, chosen before the design is fixed; the written enquiries in Section 10. Feeds: the water module, with its UV unit; the toilets module, with its fans; the market garden, with its pump, on the general circuit; the rainwater module, with its pump, if its Section 3.3 is ever built; the wood heat module, with any circulation or solar pump; the shelter, the warning and communications module, and the medical room. Takes from: the micro-hydro module, if a stream is developed.

Patterns. Grows from: not yet named. Which ground the valley keeps for hay and which may carry an array, and how a household here met the dark months before the grid, are questions for the elders. Feeds: not yet named. Resonates with: 104 Site Repair — the array, the generator, and the fuel store on poorer ground, never on the best; 252 Pools of Light — light set low and apart where people gather, dark between, which is also what the essential circuit can carry; 18 Network of Learning — the battery's record through December kept in the open, as the site's own series; 83 Master and Apprentices — paid hands learning the wiring beside the electrician who certifies it. In tension with: 105 South Facing Outdoors and 161 Sunny Place — a ground array wants the same south-facing ground as the places people sit; a roof array leaves it to them. 4 Agricultural Valleys — an array on the hayfield takes the ground the valley may hold for hay. The walk status of each is set after the Phase 1 walk, on a later date; until then, unknown (compendium Section 26).

2. Context of validity

Condition	Range this design holds for
Climate	mountain; December sun as in Section 3.2
Altitude	[set]
Group size	capacity: 34 people, the example below
Season of use	all seasons; the darkest month sizes it
Grid	none, or one that fails; Section 10
Shade	none on the array at December midday, [set]

Condition	Range this design holds for
Battery room	above 0 °C whenever the battery charges
Heat	none from electricity; the wood heat module

Where the December horizon takes the low sun, every winter figure below falls with it; the walk records the horizon before the array is placed.

3. Sizing method

What you must measure yourself before using this module:

- **Your loads** — the appliance list, with each appliance's power and hours, from datasheets, and the safety link's average draw, measured over a week with a plug-in meter, not taken from a datasheet.
- **Your sun** — the monthly figures for your region from a published source, and your December horizon, walked and recorded.
- **Your battery room** — its lowest temperature in the coldest week, with a thermometer left in it.
- **Your grid** — whether a connection exists, and what it costs to keep.

Our example: capacity, 34 people; december_sun, 0.65 to 0.74 kWh a m² a day; performance_ratio, 0.86; battery_usable, 0.90; battery_days, 4; generator_fuel, 1.0 to 1.5 L an hour; exchange_rate, 52 UAH to the euro. Values from the site parameters file, held in the workshop.

3.1 The loads

The paper's loads at 25 people, recalculated at 34: lines that grow with people are scaled by 34/25, [assumed]; fixed lines keep the paper's figure.

Load	kWh a day	Basis
Lighting, 34 points × 8 W × 5 h	1.36	scaled, [assumed]
Communications, router and phones	0.30	paper
The safety link and the shelter, together	1.50	paper
Medical refrigerator	0.50	paper
UV unit, water module	0.20	paper
Toilet fans, six, continuous	[set]	toilets module

Load	kWh a day	Basis
Essential circuit	3.86 + fans	
Kitchen refrigeration	2.18	scaled, [assumed]
Kitchen small appliances	0.54	scaled, [assumed]
Laundry	0.95	scaled, [assumed]
Laptops, documentation	0.60	paper
Workshop tools	0.50	paper
Heating circulation pump, winter	0.70	paper; 0 by gravity
General circuit, winter	5.47	
Both, winter	9.33 + fans	paper: 8, sized at 9

Loads the paper did not carry. On the essential circuit: the six toilet fans of the toilets module, 12 V, each running continuously, so that every watt a fan draws adds 0.024 kWh a day; fire detection and night lighting, and the warning receivers, when their modules are designed. On the general circuit: the market garden's pump, in the sun months, moving about 7 m³ a day in a dry week; the food store's cold room; the washroom's ventilation; the solar loop's pump in the wood heat module; and the rainwater module's pump, only if its Section 3.3 is ever built. Each is [set] until its appliance is chosen; none of the general circuit's additions runs in the darkest month but the cold room.

The heating circulation pump is zero if the wood heat module's gravity layout is built, and the winter total falls to 8.63 kWh a day.

3.2 The sun

Quantity	Calculation	Our example
Array	11 × 550 W	6.05 kWp
December, a kWp a day	0.65–0.74 × 0.86	0.56–0.64 kWh
December, the array a day	× 6.05 kWp	3.4–3.9 kWh
A year, the array	6.05 × 1,000–1,180	6,050–7,140 kWh

The December sun. The building-climate standard gives the month's sunlight on a south-facing vertical surface at three Carpathian oblast centres as 20 to 23 kWh a m², and on a horizontal one as 14 to 17, under average cloud (ДСТУ-Н Б В.1.1-27:2010, Table 8, converted from MJ; read from a scanned copy and checked against an official one before it is relied on). An array tilted at 60° receives a little more than the vertical, so the vertical is taken as its floor. That is about half the paper's 1.0 kWh a kWp a day. The stations are in the lowlands and foothills; a mountain horizon takes more.

A year. The paper's 1,000 to 1,100 kWh a kWp, and about 1,178 full-load hours for the mountain climate of the Carpathians, the lowest in Ukraine (Winkler and colleagues, Energy Conversion and Management: X 28, 2025). From March to October the array covers both circuits with a wide margin, as the paper holds; the summer surplus has no use here but the garden pump.

Not more panels. Doubling the array to chase December would still leave a deficit in the dark weeks and waste most of the summer; the battery and the generator are cheaper per December kWh. A ridge that takes the sun at two in the afternoon costs more than two panels.

Tilt and snow. About 60°, as the paper has it: near the winter optimum for 48° north, about 61° (Landau, 2017); snow barely holds on an array that steep (Andrews, Pollard, and Pearce, Solar Energy 92, 2013). The array's lower edge stands clear of the ground's snow: where shed snow piled up against it, an array lost 29 to 34 percent of its year, against 5 to 12 percent where it fell clear (Heidari and colleagues, IEEE Journal of Photovoltaics 5, 2015). The clearance is [set] from the depth of snow at the site.

3.3 The battery

Quantity	Calculation	Our example
Four days, essential	$4 \times 3.86 \text{ kWh}$	15.4 kWh, before fans
Modules, 5.12 kWh, usable	$4 \times 5.12 \times 0.90$	18.4 kWh
Days, essential alone	$18.4 \div 3.86$	4.8, before fans
Days, both circuits	$18.4 \div 9.33$	2.0
Fans the four days allow	$(4.6 - 3.86) \div 0.024$	about 31 W in all

Three modules, as in the paper, give 13.8 kWh usable, 3.6 days of the essential circuit at 34 people; the four-day rule needs four. The six fans together may draw about 31 W before a fifth module is needed: about 5 W each.

In the cold. The 48 V LiFePO₄ modules sold in Ukraine charge only between 0 and 50 to 55 °C, and discharge down to −10 or −20 °C (Deye SE-G5.1 Pro-B datasheet V2.1; Pylontech US5000 manual V1.0, 2024; Dyness BX51100). Charging below zero plates lithium in the cells: graphite and LiFePO₄

cells cycled at $-18\text{ }^{\circ}\text{C}$ lasted 185 equivalent cycles to 80 percent of their capacity, against about 2,000 at room temperature (Rauhala and colleagues, Journal of Energy Storage, 2018). The battery management of the Deye module stops charging below $0\text{ }^{\circ}\text{C}$. So the battery lives in an insulated room that stays above freezing, not in the shelter and not in an unheated outbuilding; where the room cannot be held above zero, modules with their own heating are chosen, [set].

3.4 The darkest month and the generator

Quantity	Calculation	Our example
Deficit, both circuits	$9.33 - 3.4 \text{ to } 3.9 \text{ kWh}$	5.5–6.0 kWh a day
Deficit, essential alone	$3.86 - 3.4 \text{ to } 3.9 \text{ kWh}$	0 to 0.5 kWh a day
Charge an hour, 5 kW set	the paper's figure	about 3 kWh
Running, a day	$5.5\text{--}6.0 \div 3$	about 1.8–2.0 h
Fuel, a day	$1.8\text{--}2.0 \text{ h} \times 1.0\text{--}1.5 \text{ L}$	about 1.8–3.0 L
Days the fire rules' 40 L allows	$40 \div 1.8\text{--}3.0$	about 13–22

In the darkest month the sun covers the essential circuit, or nearly so, and the battery carries the dark spells; the generator serves the general circuit. That is the paper's rule, kept: nothing critical waits on fuel. The fuel the generator burns in the darkest month is about twice the paper's, because the sun is about half. The fire rules limit what may be stored at a site to 40 L (Section 10), so fuel is bought through the winter; how much is held at any time is not published.

The generator. A 5 kW-class inverter generator, petrol, run deliberately to top the battery and held as a reserve for a long failure. A set of this class burns about 1.0 L an hour at half load and 1.5 L at three-quarters (Sakura SG6500i-A, manufacturer's figures). Petrol begins to age within about a month without a stabiliser (Honda, owner's manual); the store is turned over, and stabilised.

3.5 The inverter

A 5 kW hybrid inverter at 48 V, as the paper has it, so that a workshop tool or a washing machine starting does not trip the site; the paper's peak of about 3 kW at 25 people is [set] at 34 from the appliance list, and the garden pump's starting current is checked against the inverter before the pump is chosen. Whether the inverter is grid-tied or off-grid follows the grid question in Section 10. The rack takes a fifth and sixth module without rewiring.

4. Bill of materials

Quantities for the example. Alternatives are named in the design; none is yet known to work here.

Item	Quantity	Alternatives in the design
PV module, 550 W	11	other sizes to 6 kWp
Mounting, 60°, lower edge clear of snow	1 set	roof frame
DC cabling, breakers, surge protection	1 set	none named
Hybrid inverter, 5 kW, 48 V	1	off-grid inverter
LiFePO4 module, 5.12 kWh, 48 V	4	self-heating modules
Rack, BMS cabling, fusing	1 set	none named
Distribution board, two circuits	1	none named
The shelter's own supply	1 set	none named
Earthing and lightning protection	1 set	none named
Monitoring and metering	1 set	none named
Inverter generator, 5 kW class, petrol	1	LPG or dual fuel
Changeover or transfer switch	1	automatic transfer switch
Fuel cans, spill tray, stabiliser	1 set	none named

The array, as designed. South-facing, at about 60°, on a roof or on poorer ground; out of the December shade; its lower edge clear of the depth of snow; reachable for brushing off after a heavy fall. On the ground it is fenced; on a roof the structure is checked for snow and wind by a builder before it is fixed.

The battery room, as designed. Insulated, above 0 °C whenever the battery charges, dry, ventilated as the datasheet asks, locked, and reachable for the electrician; not in the shelter, not in an unheated outbuilding, and never warmed by an electric heater on the battery's own charge.

The shelter's own supply. The shelter keeps a small supply of its own — a battery charged from the essential circuit, with light, ventilation, and a point to charge phones — so that it works if the main system fails at the moment it is needed. It is sized from the shelter's own loads and the hours the shelter must hold; neither is published.

The generator and its fuel, as the rules ask. The generator stands outdoors, on level ground of non-combustible material, never inside a building, on a roof, on a balcony, or on an evacuation route, with its exhaust clear of doors and windows. Its fuel and oil are kept apart from it, in containers made for them, in a separate outbuilding at the distance the rules set, locked. It is wired to the building only through a changeover or transfer switch that cuts the building off from any grid. None of these is placed in any published plan.

The layout, as designed. Array, inverter and battery, distribution board, then the essential circuit and the general circuit, each on its own breakers; the shelter's supply charged from the essential circuit; the generator through its switch to the inverter. Drawn as relationships only.

5. Build

Step	Trade	Time	If done out of order
Appliances chosen, loads listed	household	[from L2]	mis-sized
Electrician chosen	household	[from L2]	no certificate
Walk: horizon, ground, room	household	[from L2]	array shaded
Enquiries answered	household	[from L2]	[from L2]
Battery room readied	builder	[from L2]	cells charged cold
Earthing, lightning protection	electrician	[from L2]	[from L2]
Mounting and array	electrician	[from L2]	[from L2]
Inverter, battery, board	electrician	[from L2]	[from L2]
Two circuits wired	electrician	[from L2]	one circuit for all
Shelter supply	electrician	[from L2]	[from L2]
Generator, switch, fuel store	electrician	[from L2]	backfeed
Measurements and certificate	electrician	[from L2]	no record

6. Cost

Estimates, not quotations: every line is the energy paper's own estimate for autumn 2026, marked "paper", scaled where the design changed. Re-pricing from dated listings was tried on 2026-10-03 and could not be done: the retail sites refused automated reading. Place: not a quotation; the paper's anchors are national. Exchange rate: exchange_rate, 52 UAH to the euro, the National Bank's mid-September 2026 rate, rounded.

The paper's anchors: a 550 W module at about 8,800 UAH; a single-phase 5 kW hybrid inverter at about 43,000 to 51,000 UAH, budget units from about 30,000; a 5.12 kWh LiFePO4 module at about 33,000 to 42,000 UAH.

6.1 Generation

Item	UAH	EUR	Basis
PV modules, 11 × 550 W	96,800	1,862	paper
Mounting frames	25,000	481	paper
DC cabling, breakers, surge protection	20,000	385	paper
Subtotal	141,800	2,727	

The mounting is the paper's; raising the array's lower edge clear of the snow may add to it, [set].

6.2 Storage and conversion

Item	UAH	EUR	Basis
Hybrid inverter, 5 kW, 48 V	48,000	923	paper
LiFePO4, 4 × 5.12 kWh	144,000	2,769	paper, scaled
Rack, BMS cabling, fusing	8,000	154	paper
Subtotal	200,000	3,846	

The battery is the paper's 108,000 UAH for three modules, 36,000 each, for four.

6.3 Distribution and protection

Item	UAH	EUR	Basis
Distribution board, with the shelter's supply	25,000	481	paper
Earthing and lightning protection	15,000	288	paper
Installation, wiring, commissioning	40,000	769	paper
Monitoring and metering	5,000	96	paper
Subtotal	85,000	1,635	

6.4 Generator

Item	UAH	EUR	Basis
Inverter generator, 5 kW class	45,000	865	paper
Changeover or transfer switch	9,000	173	paper
Fuel cans, spill tray	5,000	96	paper
Subtotal	59,000	1,135	

The separate outbuilding for the fuel, and any canopy over the generator that the enquiry allows, are not priced, [set].

6.5 Summary

Section	UAH	EUR
Generation	141,800	2,727
Storage and conversion	200,000	3,846
Distribution and protection	85,000	1,635
Generator	59,000	1,135
Construction subtotal	485,800	9,342
Design and electrician's check, 8 percent	38,900	748
Contingency, 20 percent	104,900	2,017

Section	UAH	EUR
Total external cost	629,600	12,108

Against the paper's 583,000 UAH, the difference is the fourth battery module. The percentages are the paper's own, with the contingency taken on the subtotal and the design together, as the paper does.

Contribution, not in the total. None. The labour is in the cost, paid.

Running cost.

- Generator fuel and service: the paper's 6,000 UAH a year was costed at about half the darkest month's duty found here, [set] from a dated fuel price and the first winter's log.
- Inverter: replaced at ten to fifteen years, about 50,000 UAH, paper.
- Battery: fifteen years at this cycling, then about 144,000 UAH at today's estimate, paper, scaled.
- Panels: forty years, with a slow decline in output, paper.
- Insulation and protection measured every two years, under the fire rules (Section 10), [set] from a quotation.

7. Operation and maintenance

Responsible: one named adult, paid, with a deputy, and the electrician for everything behind the board — [from L2].

- **Daily:** the battery's state read at the monitor and written down; in the darkest weeks, the general circuit shed when the battery falls to the level the household sets, [set].
- **After a heavy snowfall:** the array brushed clear from the ground, never by climbing on it.
- **Monthly:** the generator started and run under load; the fuel's date checked and the store turned over.
- **Seasonal:** before winter, the battery room's lowest temperature read; the array's shade checked against the December horizon.
- **Every two years:** insulation and protection measured by the electrician, and the record filed where inspections will ask for it.

How the programme relates to this work is described in the programme modules, in standing format.

8. Failure modes

From the design, not from experience.

Failure	Early sign	Repair and cost
Battery charged below 0 °C	capacity falls	warm room; [from L2]
Dark spell over four days	battery low	shed general; generator
Snow on the array	output near zero	brush clear from ground
Snow piled at the lower edge	array never clears	raise the array
Fuel aged	hard start	fresh fuel, stabiliser
Generator back-fed into the grid	—	switch fitted first
Fan load above 31 W	battery days fall	fifth module
Inverter tripped at a start	site goes dark	stagger the loads

9. What we got wrong

Nothing yet: this module has not been built.

10. Regulatory notes

Country	Instrument, number, date	Checked by, date
Ukraine	found 2026-10-03, not checked — see below	—
Romania	not checked	—
Poland	not checked	—
Slovakia	not checked	—
Hungary	not checked	—
Serbia	not checked	—
Czechia	not checked	—

Ukraine, found and not checked. Read on 2026-10-03 on secondary copies — LIGA:ZAKON, a university copy of the fire rules without their later amendments, and the State Energy Supervision Inspectorate's own pages — because the statute site could not be reached; each is read on zakon.rada.gov.ua and signed before this row counts as checked.

- **Fire Safety Rules of Ukraine, Order No. 1417 of the Ministry of Internal Affairs, 30.12.2014**, amended last by Order No. 67 of 27.01.2026. Section IV, chapter 1, point 1.20: insulation resistance and short-circuit protection measured at least every two years, unless the technical operating rules set another interval. Section III, chapter 2, points 2.14 and 2.18: no store of flammable or combustible liquids in residential, public, or administrative buildings.
- **Order No. 474 of the Ministry of Internal Affairs, 11.07.2024**, in force 14.08.2024, adding to the fire rules a chapter on generators of up to 20 kW with a tank of up to 100 L: outdoors, on level ground of non-combustible material; never in a building, on a roof, a balcony, or an evacuation route; fuel and oil of at most 40 L and 10 kg, in a separate outbuilding at least 7 m from buildings; no generator plugged into a socket. Order No. 67, 27.01.2026, in force 27.02.2026, adds rules for groups of generators.
- **Rules for the Technical Operation of Consumers' Electrical Installations, Order No. 258 of the Ministry of Fuel and Energy, 25.07.2006**: for a private owner of an installation up to 1 kV, they apply only to measuring the wiring's insulation; results are kept in protocols. The intervals sit in COV-H EE 20.302:2007, not read.
- **Fire Safety Rules for the Education System, Order No. 974 of the Ministry of Education and Science, 15.08.2016**, Section V, points 9 and 11: measurements as the general fire rules ask; hand torches on batteries for every member of the night duty where people stay round the clock. Whether the site falls under these rules depends on its legal form.
- **Law No. 2019-VIII, on the electricity market**, amended last by Law No. 4963-IX of 02.09.2026, and **Law No. 3220-IX of 30.06.2023**: an active consumer may produce, store, and sell surplus; self-generation is open to household solar of up to 30 kW and small non-household consumers of up to 50 kW. No duty was found to register an installation with no grid connection at all.
- **The Distribution System Code, NEURC Resolution No. 310 of 14.03.2018**, amended last by No. 1355 of 11.08.2026: governs the connection of a consumer's generation; not read.
- **The State Energy Supervision Inspectorate**, guidance of 2023 and 2024: a generator reaches the building's wiring only through a changeover or automatic transfer switch after the meter and the main breaker, so that the building is cut off from the grid, and the connection is made by a specialist. The State Emergency Service's guidance of 2025 and 2026, as reported, keeps a generator at least 6 m from doors and windows and asks for a carbon monoxide detector in living rooms.
- **Not reached**: the Rules for the Arrangement of Electrical Installations, 2017; ДСТУ HD 60364-6; the standard for lightning protection; the standard for emergency lighting.

One correction to the paper. The paper expects an annual electrical inspection of an establishment that receives children. No instrument found asks for one; the fire rules ask every two years. The enquiry asks.

The written enquiries: to the regional emergency service, the generator's place, the fuel store, and the interval of measurements for this establishment; to the distribution operator, if a grid connection exists, the inverter, the changeover, and whether any notice is due; to the electrician, which standards the certificate will be signed against.

11. Adaptation notes

The design is a seed, not a copy: a different place grows a different form from it.

- **A sunnier valley:** the December figure rises, the generator runs less; keep the four days.
 - **A darker valley, or a deep horizon:** the generator runs more, or a stream carries the winter; never more panels first.
 - **A colder valley:** the battery room is the hard part; self-heating modules, or the battery inside the heated envelope.
 - **A smaller group:** recalculate Section 3.1; the battery keeps four days of the essential circuit.
 - **A larger group:** above about 31 W of fans, or more essential load, a fifth module before more panels.
 - **A reliable grid:** the battery may shrink to the outages the grid has; the essential circuit and its four days stay for an establishment that receives children.
 - **A stream:** the micro-hydro module.
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12. Provenance and licence

- **Written by:** Michel Garand. [set — others as they choose to be named]
- **From whose knowledge:** the energy working paper, EN v0.1; ДСТУ-Н Б В.1.1-27:2010; Winkler and colleagues, 2025; PVGIS 5; Landau, 2017; Andrews, Pollard, and Pearce, 2013; Heidari and colleagues, 2015; the Deye, Pylontech, and Dyness datasheets; Rauhala and colleagues, 2018; Sakura and Honda manufacturer data; the fire rules and the other instruments in Section 10, as read on secondary copies.
- **Traditional knowledge:** none. How the valley met the dark months is for the elders.
- **Text:** CC BY-SA 4.0
- **Drawings and designs:** [hardware licence — open decision]
- **Measurement data:** none yet

- **Gate 1:** household decision, 2026-10-03, recorded by Michel Garand — v0.1 may be published as it stands, at L1, in English, Ukrainian, and German
 - **Wartime review:** 2026-10-03, Michel Garand — v0.1 read against compendium Section 2; nothing places the site; the shelter, the safety link, the battery, the generator, and the fuel store in standing format; accepted
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13. What to check

- **The appliance list** — every load in Section 3.1, replacing the scaled lines.
 - **The safety link** — its average draw, measured over a week.
 - **The fans** — their power, against the 31 W the battery allows.
 - **The December sun** — the standard's Table 8 read in an official copy; the site's December horizon walked.
 - **The grid** — whether a connection exists, and the Distribution System Code read for it.
 - **The battery room** — its lowest temperature in the coldest week; whether self-heating modules are needed.
 - **The snow** — its depth at the site, which sets the array's clearance.
 - **The enquiries** — the generator's place, the fuel store, the interval of measurements.
 - **Quotations** — every line in Section 6, dated, to replace the paper's estimates; a dated fuel price for the running cost.
 - **The Ukrainian row** — every instrument read and signed.
 - **From L2, once built:** build times and out-of-order consequences in Section 5; the maintenance record in Section 7; early signs and repair costs in Section 8; Section 9.
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